

Module 20: Behavioral & Budget Leakage Patterns Explanation Guide

This document details the behavioral and statistical anomaly detection methodologies applied to the ECHS database. It maps clinical policies, real-world examples, and backend database checking rules for the identified budget leakage vectors.

1. CORPORATE & CATEGORICAL OVERBILLING

These patterns detect systematic overcharging and package billing inflation concentrated by facility category or corporate groups.

A. Corporate Hospital Overbilling (Pattern 1)

Kaise Fraud hota hai: Empanelled private hospital chains ya specific high-volume single hospitals system ko abuse karte hain by systematically overbilling ECHS. Yeh surgeries, procedures, medicines aur clinical packages ke rates ko agreement limits se bahut zyada dikha kar bill submit karte hain.

Hindi Example: Ek private hospital chain (jaise Park Hospital) standard treatments ke rates artificially badha deti hai aur package pricing limits ko cross karti hai. ECHS audits ke dauran inka ₹437 Crores deduct (reject) kiya jata hai. Vijay Hospital [ID 3149] me ₹149 Crores ka claim reject hua kyunki unka deduction rate abnormal (34.00%) tha.

Policy Violation: Empanelment MoU and pre-agreed ECHS tariff package rates guideline ka direct violation.

Database Check: Hospital leakage summary dataset check kiya jahan unique hospital name with ID code (hospital_name_with_id) ko absolute deductions amount (total_deducted_lakh) aur percentage deduction (deduction_pct) ke hisab se sort aur rank kiya.

B. Facility Category Leakage — Type M & N (Pattern 2)

Kaise Fraud hota hai: ECHS database me hospital classification structures hote hain. Type M (Military Establishments empanelled) aur Type N (Naval / Specialist Facilities) me normal system average (~9.96%) se double (25.96% aur 23.39%) deduction rate dekhne ko mila, jo standard package codes ki galat mapping ya manual overrides ko darshata hai.

Hindi Example: Ek specialty clinic (Type N Category) standard package codes ki jagah high-end surgical procedures ke galat code mapping submit karti hai, jiske chalte claims process check ke dauran automatic billing engine 23% se zyada amount reject kar deta hai.

Policy Violation: ECHS Office Master coding protocols aur category-wise treatment approval rules ka violation.

Database Check: Table metadata me check kiya jahan hosp_type_code ki value 'M' ya 'N' thi, aur total_deducted_cr ko overall category wise check kiya: $(total_deducted_cr / total_claimed_cr) * 100$ double the system average cross kar raha tha.

C. Quality Accreditation Compliance Gap (Pattern 3)

Kaise Fraud hota hai: Non-NABH (unaccredited) hospitals ke billing standards aur quality checkpoints behtar nahi hote. Is wajah se non-accredited private hospitals ka leakage / deduction rate (9.87%) accredited hospitals (8.24%) se significantly high hota hai, jo system me billing non-compliance aur excessive errors ko lead karta hai.

Hindi Example: Ek non-NABH hospital billing rules aur required clinical documents ki checksheets ignore karke claims submit karta hai, jiske karan unke bills me redundant entries catch hoti hain aur unka audit rejection rate badh jata hai.

Policy Violation: Quality assurance guidelines aur standard clinical documentation compliance protocols.

Database Check: Summary details me hospital quality status (`nabh_status = 'N'`) wale claims ko analyze kiya aur unhe accredited (`nabh_status = 'Y'`) claims ke percentage deduction (`deduction_pct`) ke sath directly compare kiya.

2. BEHAVIORAL EXPLOITATION & BED ABUSE

Hospitals use these behavioral strategies to artificially inflate length of stay or claim counts for the same patient.

A. Extended Length of Stay (LoS) Bed Blocking (Pattern 4)

Kaise Fraud hota hai: Hospital patient ko fully recover hone ke baad bhi discharge karne ki bajaye extra days tak ward/room me admit rakhta hai (bed blocking), taaki daily room rent, doctor rounds, aur monitoring charges ECHS se extra claim kar sake.

Hindi Example: Ek patient ka simple hernia surgery hota hai jise 3 din me discharge ho jana chahiye tha. Lekin hospital patient ko 12 din admit rakhta hai bina kisi clinical complication ke, aur daily room rent aur service fees bhejta rehta hai.

Policy Violation: Clinical Necessity criteria aur Maximum permissible Length of Stay (LoS) guidelines.

Database Check: Table `claim_intimation` ke admission date (`CI_ADMISSION_DATE`) aur `claim_submission` ke discharge date (`CS_DOD`) ke bich date difference: `DATEDIFF(CS_DOD, CI_ADMISSION_DATE) > 10`.

B. Ping-Pong Admissions (Pattern 5)

Kaise Fraud hota hai: Hospital single surgery package limits bypass karne ke liye patient ko discharge karta hai aur 48 ghante ke andar fir se admit dikha deta hai taaki do alag-alag treatment package payouts claim kar sake.

Hindi Example: Patient ko appendicitis aur urinary stone dono ka treatment chahiye. Hospital ek hi admission me dono operation karne ke bajaye, appendix removal karke discharge karta hai aur next day hi stone operation ke liye fir se admit dikha deta hai, taaki do alag package payouts claim kar sake.

Policy Violation: Package Splitting and Unbundling rules under ECHS guidelines.

Database Check: Window function LEAD use karke patient (`CI_SERVICE_NO` and `CI_BENEFICIARY_NAME`) ke sequential admissions check kiye: `DATEDIFF(next_admission_date, discharge_date) BETWEEN 0 AND 2`.

C. ICU Conversion Abuse (Pattern 6)

Kaise Fraud hota hai: High room rents aur package values collect karne ke liye hospitals stable patients ko general/private rooms ki jagah fake ICU monitoring requirements dikha kar ICU beds me admit kar dete hain.

Hindi Example: Routine post-surgery checkup ke patient ko (jise normal private room ki zaroorat hai) hospital extra financial gain ke liye emergency ICU zone me transfer kar deta hai taaki ECHS se ₹15,000/day ka ICU package rate bill kiya ja sake.

Policy Violation: Clinical eligibility criteria for intensive care (ICU) admissions.

Database Check: Database me `CS_ROOM_TYPE = 'ICU'` percentage metrics monitors check kiye: `(icu_cases * 100 / total_claims) > 15%`.

3. SYSTEM CHECK & CONTROL CIRCUMVENTION

Hospitals exploit calendar timings, surgeon credentials, and financial delegation limits to bypass audit checking protocols.

A. Weekend / Holiday Surge Admissions (Pattern 7)

Kaise Fraud hota hai: Hospital planned / elective procedures ko jaan-बूझकर Friday night, Saturday ya Sunday ko admit dikhate hain jab regional referral polyclinic staff closed/skeleton staff mode me hota hai. Aisa karke wo normal screening checks ko bypass karke direct emergency bypass use kar lete hain.

Hindi Example: Patient ki planned cataract surgery honi thi, par hospital use Saturday ko admit dikhata hai emergency route se taaki polyclinic referral verification officer bill bypass karke seedhe bill submission clear kar sake.

Policy Violation: Out-of-hours admissions referral guidelines aur emergency admission verification checks.

Database Check: MySQL function DAYOFWEEK(CI_ADMISSION_DATE) IN (1, 6, 7) check kiya (Sunday=1, Friday=6, Saturday=7) aur un hospitals ko highlight kiya jinka weekend admission levels abnormal level par the.

B. Superman Surgeon / Doctor Cloning (Pattern 8)

Kaise Fraud hota hai: Hospital claims portal par single main specialist surgeon ke name par ek hi din me clinically impossible number of surgeries (jaise 15-20 surgeries) document kar deta hai. Asal me junior staff surgeries kar rahe hote hain, ya fir paper-only admissions hote hain.

Hindi Example: Ek cardiologist ke profile se ek hi din me 18 complex bypass surgeries billed dikhai jati hain. Ek surgeon ke liye single day me itni surgeries karna namumkin hai. Asal me junior staff ne kiya ya documents fake banaye gaye.

Policy Violation: Surgeon credential verification standards aur clinical practice limits.

Database Check: Treatment doctor (CS_TREAT_DOCT) aur admission date (DATE(CS_SUB_DATE)) par grouping lagai aur filters apply kiye jahan clean entries count COUNT(*) >= 15 per day perform kar rahi thin.

C. Threshold Avoiding — The ₹99k Trick (Pattern 9)

Kaise Fraud hota hai: ECHS rules ke anusaar ₹1 Lakh se bade bill hone par pre-authorisation higher authorities (CFA) verify karte hain. Is checking zone se bachne ke liye hospitals intentionally bills ko ₹99,000 se ₹99,999 ke range me limit kar dete hain taaki direct billing release ho jaye.

Hindi Example: Gallstone surgery ka actual bill ₹1,15,000 banana chahiye tha, par hospital use janbujhkar system limits ke niche ₹99,850 ka banata hai taaki dynamic pre-approval checks automatic pass-through ho jayein.

Policy Violation: Delegation of Financial Powers Rule (DFPR) bypass and split-billing evasion code.

Database Check: Submitted claim values CS_NET_CLAIM_AMT BETWEEN 99000 AND 99999 with count COUNT(*) > 10 per hospital.

"Module 20 proves that budget leakage within ECHS is driven by systemic overbilling, bed-blocking, and intentional bypass of financial thresholds. The findings provide the ECHS Directorate with direct data-driven policies to enforce NABH compliance, audit weekend admissions, and automatically flag unbundled ping-pong claims."